Spur Gear and Gear Train Design

CAD Mechanical Design Portfolio

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Developed Using

Fusion360

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Table of Contents

1 PROJECT OVERVIEW

This project focused on the design, assembly, and analysis of a spur gear pair using Autodesk Fusion 360. The objective was to develop an understanding of gear geometry, gear ratios, motion transmission, and mechanical assemblies. A two-gear spur gear train consisting of a 20-tooth driver gear and a 40-tooth driven gear was created and assembled to demonstrate speed reduction and torque multiplication.

The project also involved generating engineering drawings, applying motion relationships, and creating realistic renders suitable for a CAD engineering portfolio.

2 GEAR THEORY

Gears are mechanical components used to transmit rotational motion and power between shafts. Spur gears are the most common type of gear and consist of straight teeth mounted parallel to the shaft axis.

2.1 ADVANTAGES OF SPUR GEARS

- High efficiency
- Simple manufacturing
- Accurate speed transmission
- Reliable operation

2.2 COMMON APPLICATIONS

- Robotic systems
- Gearboxes
- Automotive transmissions
- CNC machines
- Industrial machinery

When two gears mesh, the rotational speed and torque depend on the number of teeth on each gear.

3 GEAR PARAMETERS

3.1 DRIVER GEAR (GEAR_20T)

Parameter	**Value**
Number of Teeth	20
Module	2 mm
Pressure Angle	20°
Pitch Diameter	40 mm
Outside Diameter	44 mm
Thickness	10 mm
Bore Diameter	10 mm
Root Fillet Radius	0.6 mm

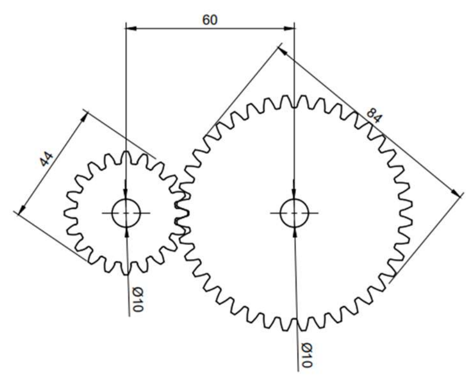
3.2 Driver Gear (Gear_40T)

Parameter	Value
Number of Teeth	40
Module	2 mm
Pressure Angle	20°
Pitch Diameter	80 mm
Outside Diameter	84 mm
Thickness	10 mm
Bore Diameter	10 mm
Root Fillet Radius	0.6 mm

3.3 Assembly Parameters

Parameter	Value
Center Distance	60 mm
Gear Ratio	2:1

4 CAD Modelling



The gears were generated using the Spur Gear Add-In available within Fusion 360. Parametric gear generation allowed the tooth geometry to be created automatically based on the specified module, pressure angle, and number of teeth.

The modeling process included:

1. Generating a 20-tooth spur gear.
2. Generating a 40-tooth spur gear.
3. Creating shafts for mounting.
4. Positioning gears at the calculated center distance.
5. Applying revolute joints.
6. Creating a motion-linked assembly.
7. Producing engineering drawings and renders.

The parametric approach allows the design to be modified easily by changing the gear parameters.

5 GEAR RATIO ANALYSIS

The gear ratio is determined using:

$$Gear\ Ratio = \frac{Driver\ Gear\ Teeth}{Driver\ Gear\ Teeth}$$

Substituting the values:

$$Gear\ Ratio = \frac{40}{20}$$

$$Gear\ Ratio = 2$$

Therefore:

$$Gear\ Ratio = 2 : 1$$

5.1 SPEED ANALYSIS

$$Output\ Speed = \frac{Input\ Speed}{2}$$

Example:

Input Speed = 100 rpm

Output Speed = 50 rpm

5.2 TORQUE ANALYSIS

$$Output\ Torque = Input\ Torque \times 2$$

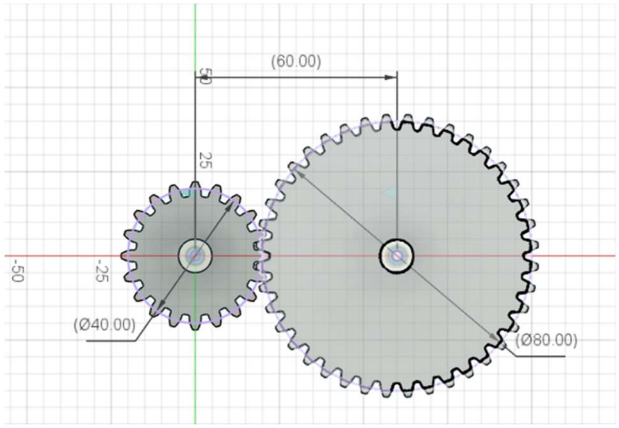
Example:

Input Torque = 1 Nm

Output Torque = 2 Nm

This demonstrates the trade-off between speed and torque commonly used in mechanical power transmission systems.

6 ASSEMBLY



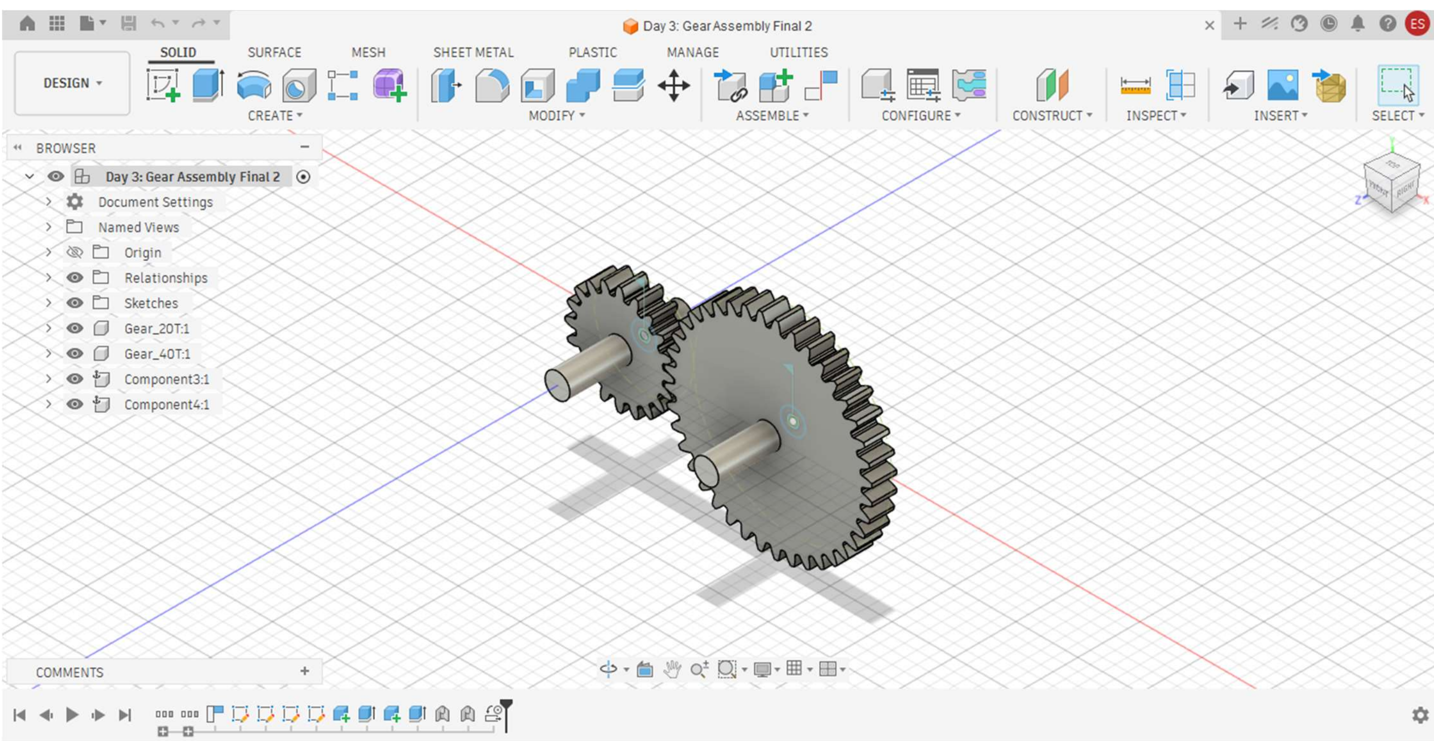
The gear train assembly was created by positioning the gears at a center distance of 60 mm.

Two shafts were modeled and used to support the gears. Revolute joints were applied to allow independent rotation about each shaft axis. A motion link was then established between the gears to simulate realistic gear meshing.

The assembly successfully demonstrated:

- Rotational motion transfer
- Opposite direction rotation
- Speed reduction
- Torque multiplication

7 MOTION SIMULATION



Motion simulation was performed using Fusion 360 motion relationships.

The driver gear was rotated manually while the driven gear responded automatically according to the specified gear ratio.

Observed behavior:

- Driver Gear Rotation: 1 Revolution
- Driven Gear Rotation: 0.5 Revolution
- Rotation Direction: Opposite
- Motion Transfer: Successful

The simulation verified the theoretical gear ratio calculations.

8 RESULTS

The project successfully achieved all design objectives.

Parameter	Result
Driver Teeth	20
Driven Teeth	40
Gear Ratio	2:1
Center Distance	60 mm
Speed Reduction	50%
Torque Increase	100%

The assembly functioned correctly and demonstrated realistic mechanical behavior.

9 APPLICATIONS

Spur gear trains are widely used in engineering systems requiring speed reduction and torque amplification.

Applications include:

- Robotic joints
- Mobile robots
- Industrial gearboxes
- Conveyor systems
- Machine tools
- Automotive power transmission systems
- CNC equipment

The concepts learned in this project are directly applicable to robotics and mechatronics engineering.

10 Future Improvements

Several enhancements can be made to the current design:

- Add bearings for shaft support.
- Create a multi-stage gear train.
- Perform finite element stress analysis.
- Investigate gear tooth contact stresses.
- Design a gearbox housing.
- Explore helical gear alternatives.
- Conduct dynamic motion studies.

These improvements would increase the realism and engineering depth of the project.

11 Conclusion

A complete spur gear pair and gear train assembly were successfully designed and analyzed using Autodesk Fusion 360. The project demonstrated fundamental mechanical engineering concepts including gear geometry, speed reduction, torque multiplication, assembly design, and motion simulation. The resulting model, engineering drawings, and rendered images form a valuable addition to a mechanical design and robotics portfolio.